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TECHNICAL NOTE

Effects of Wetting and Cyclic Wetting–Drying on Tensile Strength of Sandstone with a Low Clay Mineral Content

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1 Introduction

Sandstones are encountered in many underground excavations and mines, and in addition, because of their abundance, they are commonly used as building stone and aggregate material for dams, railway or airport foundations, and other infrastructures (Shakoor and Barefield 2009). Therefore, understanding the mechanical behavior of sandstones and how they are affected by environmental factors such as saturation or temperature is clearly important. A number of experimental studies have shown that various sandstones could suffer a significant loss of uniaxial and confined compressive strength upon saturation (Colback and Wild 1965; Burshtein 1969; Hawkins and McConnell 1992; Vasarhelyi 2003; Lin et al. 2005; Li et al. 2005; Aydan 2012; Duda and Renner 2013; Wasantha and Ranjith 2014). For clay-bearing rocks, the saturation-induced reductions in strength and stiffness can be up to 80 % (Cherblanc et al. 2016; Vergara and Triantafyllidis 2016). In contrast, Reviron et al. (2009) did not observe any notable effect of water on the strength of sandstones composed almost exclusively of quartz with the absence of clay minerals. Hawkins and McConnell (1992) showed that the sudden decrease in strength commonly occurred at a moisture content well below saturation. Wong and Jong (2014) found that the tensile strength of an artificially molded gypsum rock called Hydrocal B-11 dropped to nearly half of its dry value after being soaked in water for one week; the tensile strength continued to fall slightly after the samples were immersed in water for three and ten weeks. However, how the tensile strength of sandstone changes when it has been soaked in water for a long period is still not well understood.

In addition to long-term wetting, dams and infrastructure foundations may also be subjected to cyclic wetting–drying during their construction and operation, owing to rain or changes in groundwater levels. Cyclic wetting–drying is another acknowledged yet poorly understood rock weathering process. The physical and mechanical properties of sandstone samples deteriorate to different degrees after cyclic wetting–drying treatments (Table 1). Hale and Shakoor (2003) reported no observable effect on the uniaxial compressive strength of sandstone even after 50 wetting–drying cycles. However, a recent study by Deng et al. (2012) showed that cyclic wetting–drying can significantly decrease the mechanical properties of sandstone. In the studies cited, the samples were tested under wet condition after the cyclic wetting–drying treatments, but their properties were not systematically investigated under dry condition after the samples experiencing wetting–drying cycles. Therefore, the effect of cyclic wetting–drying on sandstone's tensile strength still remains incompletely understood.

This experimental study was focused on the following two issues: (1) the effect of immersion time on sandstone deterioration along with the recovery of tensile strength from saturated to dry conditions and (2) the effect of cyclic wetting–drying on sandstone tensile strength.

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Table 1 Experimental studies on the effects of cyclic wetting–drying on the mechanical properties of various sandstones

References	Rock type	Sample size	Wetting and drying methods	Compressive strength	Tensile strength
Hale and Shakoor (2003)	Six sandstones from northeastern Ohio and western Pennsylvania	ϕ5.4 cm × 10.8 cm	Wetting: immersed in water for 1 day Drying: oven-dried at 110 °C for 1 day Testing: under dry condition	No observable reduction	
Lin et al. (2005)	Tertiary sandstones	ϕ5.5 cm × 12.5 cm	Wetting: immersed in water for 1 day Drying: not addressed with details Testing: under dry condition	Decrease or no effects	
Deng et al. (2012)	Sandstone	ϕ5 cm × 10 cm	Wetting: immersed in water from Yangtze river for 30 days under different water pressures Drying: air-dried for 5 days Testing: under wet condition	Decrease	
Khanlari and Abdilor (2015)	Sandstone	ϕ5.4 cm × 10.8 cm	Wetting: immersed in water for 1 day Drying: oven-dried at 110 °C for 1 day Testing: under dry condition	No observable reduction	
Hua et al. (2015)	Sandstone	Φ7.5 cm × 2.5 cm	Wetting: immersed in water for 2 days Drying: oven-dried at 105 °C for 1 day Testing: under wet condition		Decrease

2 Experimental Method

2.1 Sample Preparation

A total of 48 samples were cored from an intact sandstone block, collected from a construction site at the Jiangbei International Airport, Chongqing, China. A modal analysis of three thin sections showed that the sandstone was composed of about 50 % quartz, 28 % feldspar, 20 % rock fragments and ≤2 % clay minerals (kaolin and illite) in volumetric fraction. Rock fragments mainly consisted of siliceous rock, mudstone and phyllite. The sandstone grains were angular or subangular, and they ranged between 0.1 mm and 0.2 mm in size. The sandstone block from which the samples were cored was fresh and massive; and the average porosity was about 12 %. Overall, the investigated sandstone is classified as a fine-grained feldspathic arenite containing about ≤2 % clayey matrix and little calcite cement. According to the recommendations of the International Society for Rock Mechanics (ISRM 1978), the sandstone samples used for the tensile strength tests were disk-shaped with a diameter of 5 cm and a thickness of 2.5 cm. The samples were stored in dry air at room temperature before testing.

2.2 Test Procedure

Two sets of weathering conditions were considered, continuous wetting and cyclic wetting–drying. For the continuous wetting tests, the saturated sandstone samples were immersed in water for different lengths of time (16, 24, 40 and 60 days), followed by Brazilian tensile testing under wet condition. Note that during the tests, the samples failed within two minutes of the load being applied, and thus it was assumed that water loss from the samples during testing due to either evaporation or expulsion was negligible. That is, the degree of saturation was regarded as approximately constant during the tests. For the cyclic wetting–drying tests, Brazilian tensile tests were performed after the samples experienced 2, 4, 6, 10 and 15 wetting–drying cycles.

The approaches and the immersion or dry times used for generating wetting–drying cycles differ considerably in the literature (Table 1). Aydan and Ulusay (2003) and Sato and Aydan (2014) showed that the diffusion of water into (or out of) a sample requires an appropriate duration. For the experiments described in this paper, a number of preliminary tests were conducted to examine the length of time required for samples to reach a constant mass during wetting and drying processes, respectively. The samples were wetted in the following four steps: (1) the lower one-

quarter thickness of the sample was soaked in water for 2 h; (2) the lower half of thickness of the sample was soaked in water for 2 h; (3) the lower three-quarters thickness of the sample was soaked in water for 2 h; 4) water was added to make the entire sample fully soaked in water (Fig. 1). Once the samples became fully saturated, their mass would stay approximately constant. The small fluctuations in mass during the saturation stage were due to the water adsorbed by the sample surface. After the samples were soaked in water for about 18 h, they were removed from the water and allowed to dry in air at room temperature. During the drying stage, the rate of mass decrease gradually dropped as time passed, and no reductions in mass could be seen after the samples had been dried for about 3 days. Based on these tests, the wetting–drying cycle times for this study were defined as soaking the samples for at least 18 h and then drying the samples in air for 3 days. A Keyence VHX-600 digital microscope (Keyence Corporation, Osaka, Japan) was used to examine the minerals, texture, cementation and alteration after the wetting–drying cycles.

In order to reduce the discreteness of the experimental data, five samples were subjected to each of the above treatments. Three untreated samples were also tested to determine the initial Brazilian tensile strength of the sandstone without water weakening. Brazilian tensile tests were performed under wet condition for the continuous wetting samples and under dry condition for the wetting–drying samples, respectively. The procedure was designed to test the strength recovery from a saturated to a dry state. A vertical loading rate of 0.1 kN/s was applied to the samples. A high-speed video camera (i-SPEED TR, Olympus, Tokyo, Japan) was used to record the failure of sandstone samples during Brazilian tensile testing in order to document the influence of water weakening on failure.

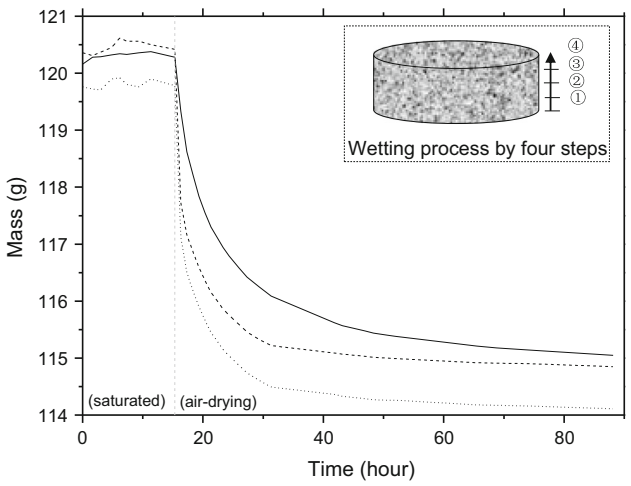


Fig. 1 Mass changing of three sandstone samples during the wetting and drying processes of the preliminary tests

3 Experimental Results

3.1 Continuous Wetting

Sandstone tensile strength after a soaking for 16 days dropped to about 59 % of the initial value of tensile strength under dry state (Fig. 2; Table 2). This finding is similar to that of Burshtein (1969), who found that tensile strength was sensitive to moisture content. However, sandstone samples that were immersed in water for longer periods did not show any significant additional reduction in tensile strength. Duda and Renner (2013) also found that storing samples in distilled water for different time periods did not systematically affect the strength of the sandstone samples at a strain rate of 10^{-5} s^{-1}.

3.2 Cyclic Wetting–Drying

The general trend shown in Fig. 2 is that cyclic wetting–drying tends to decrease the sandstone's tensile strength, but some fluctuations exist that partially obscure this trend. The samples experiencing only two wetting–drying cycles exhibited the highest average tensile strength, 5.28 MPa, but after 15 wetting–drying cycles, the average tensile strength was reduced to 4.72 MPa (Table 2). This is, however, only a 5 % reduction in tensile strength compared with the initial tensile strength of the dry sandstone. These results are similar to those of Hale and Shakoor (2003), Özbek (2014) and Khanlari and Abdilor (2015). Those researchers also did not find an obvious reduction in unconfined compressive strength due to cyclic wetting–drying treatment, although they pointed out that the possible small strength reductions could be masked by the inter-sample variability inherent in the strength

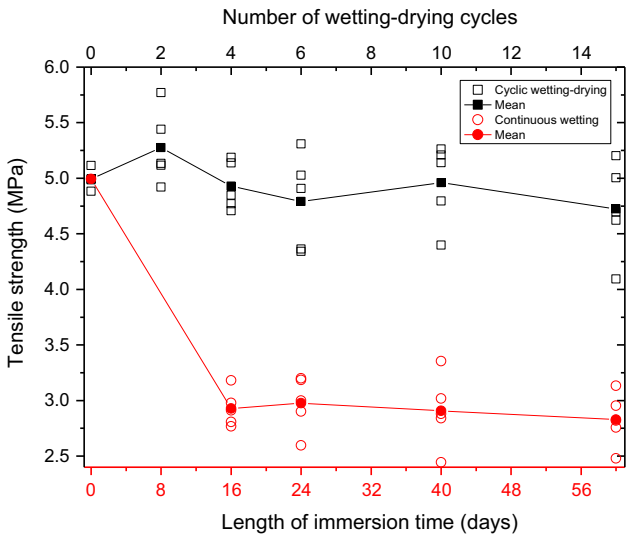


Fig. 2 Changes in Brazilian tensile strengths under continuous wetting and cyclic wetting–drying conditions

Table 2 Experimental results of the effects of continuous wetting and cyclical wetting–drying on the sandstone's tensile strength

Scenarios	Tensile strength (MPa)							$\sigma_t/\sigma_{t(\text{initial})}$
	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Average	Coefficient of variation	
Initial								
0	4.88	4.98	5.11			4.99	0.02	1.00
Length of immersion time (days)								
16	2.77	2.98	3.18	2.81	2.91	2.93	0.06	0.59
24	3.19	3.20	3.00	2.90	2.60	2.98	0.08	0.60
40	2.44	3.02	3.35	2.84	2.88	2.91	0.11	0.58
60	2.95	2.76	2.48	3.13	2.82	2.83	0.09	0.57
Number of wetting–drying cycles								
2	5.77	5.44	5.13	4.92	5.12	5.28	0.06	1.06
4	5.19	5.14	4.77	4.85	4.71	4.93	0.04	0.99
6	4.91	4.34	4.36	5.31	5.03	4.79	0.09	0.96
10	4.40	5.21	4.79	5.26	5.14	4.96	0.07	0.99
15	4.09	5.00	4.70	5.20	4.62	4.72	0.09	0.95

measurements. The issue of inter-sample variability also exists in this study; it is discussed later in Sect. 4.2. Lin et al. (2005) observed that sandstone with 11 % chlorite lost 20 % of its strength after 60 wetting–drying cycles, but this loss of strength was not observed in a sandstone with 1 % chlorite. This indicates that sandstone mineralogy may determine its sensitivity to cyclic wetting–drying treatments.

3.3 Comparison between the Two Weathering Conditions

The most obvious difference between continuous wetting and cyclic wetting–drying was that a more significant reduction in tensile strength occurred during continuous wetting. The reason appeared to be that the samples experiencing continuous wetting were tested under wet conditions. The reduced influence of cyclic wetting–drying on tensile strengths indicates that much of the tensile strength lost during wetting can be recovered once the samples dry. Baechle et al. (2005) studied elastic weakening related to water adsorption and found that the weakening was reversible and disappeared after drying. A common thread between the two sets of weathering experiments run for this study was that the coefficients of variation were larger for samples that had been immersed in water for longer periods of time or experienced a greater number of wetting–drying cycles (Table 2). This is different than the results reported by Wong and Jong (2014). In their experiments, the tensile strength values obtained for wet samples were less scattered than those for dry samples.

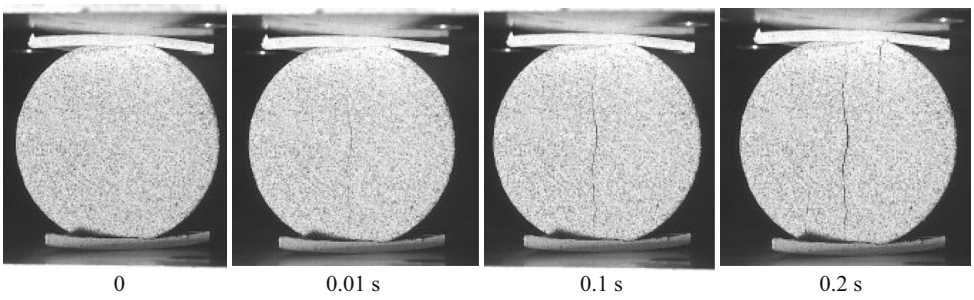
Figure 3 shows the processes of crack development associated with failure for samples subjected to Brazilian tensile testing. A primary crack was initiated at the center of the sample and then propagated vertically along the loaded diameter. This sequence was the same for both samples that had undergone continuous wetting and samples subjected to cyclic wetting–drying. The difference between the two sets of samples was crack propagation speed as determined from analysis of the high-speed video images. For dry samples, it took around 0.2 s for the primary crack to propagate from the center to the loaded ends, whereas the primary crack propagation took nearly 5 s in wet samples. During Brazilian tensile testing, a loud of "boom" was heard when dry samples broke, but wet samples failed with little noise. This observation indicates that the presence of water substantially reduced the sandstone's brittleness.

4 Discussion

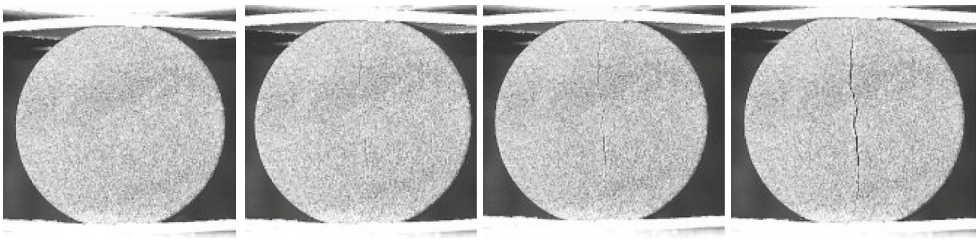
4.1 Weakening Mechanisms

Van Eeckhout (1976) reviewed the possible mechanisms responsible for saturation-induced strength reduction in rocks. These are summarized briefly below.

1. Fracture energy reduction: The presence of water within the micropores or cracks of a rock can lower the surface free energy, which in turn facilitates further crack propagation and reduces the rock's strength.
2. Capillary tension decrease: The capillary tension is a function of the vapor pressure in equilibrium with capillary water, and filling the micropores or cracks of a rock with water can reduce the capillary action.
3. Pore pressure increase: For rocks of low permeability or high loading rates, bulk compaction due to reduction



(a) Dry sample



(b) Wet sample

Fig. 3 Cracking process of dry and wet samples

in pore volume leads to an increase in pore pressure and, as a consequence, to a decrease in effective stress, which may ultimately cause a reduction in strength.

4. Frictional reduction: The friction coefficient at grain contacts may decrease in the presence of water. In particular, the friction coefficient of clay minerals/cements saturated with water may drop to 40–80 % of the value under dry conditions.
5. Chemical and corrosive deterioration: Softening and possible expansion of the clay minerals can contribute to the strength reduction in the presence of water. Dissolution of cementing material, constituent mineral grains, or both, in some cases associated with re-precipitation of dissolved ions as salts, comprises a set of chemical reactions that can occur during cyclic wetting–drying.

As Van Eeckhout (1976) discussed, the above five processes may occur simultaneously during the deformation of wet rocks, but some of them will be more significant than others for certain rock types under certain conditions. Baud et al. (2000) showed that the lower brittle strength of sandstones in the presence of water is mainly due to reductions in both fracture energy and friction coefficient. These factors may also contribute to the tensile strength reduction in wet sandstone samples in this study. Capillary tension decrease is also one of the possible reasons for the strength reduction in wet sandstone samples used in the experiments. The experiments used a loading rate of 0.1 kN/s, so weakening of the samples by the development of transient high pore fluid pressures seems rather unlikely because the Brazilian tensile tests were performed under "drained" conditions. It is possible that chemical and corrosive deterioration played a role in lowering the tensile strength of the wet sandstone samples, but these factors were probably minor because clay minerals were not abundant in the sandstone tested. In addition, tensile strength and immersion time were nearly independent, indicating that continuous dissolution of cementing material or mineral grains is unlikely. Duda and Renner (2013) showed that the magnitudes of physicochemical water weakening were moderate and similar for the two clay-bearing sandstones and the pure quartz arenite they studied, which also demonstrates the minor role of chemical and corrosive deterioration in weakening sandstones. Pimienta et al. (2014) observed that elastic weakening in a quartz–pure sandstone increased with porosity and argued that weakening from moisture does not originate from the presence of clay minerals but from the rock microstructure. Therefore, it may be concluded that reductions in both fracture energy and friction coefficient are the most important mechanisms for saturation-induced reduction in the tensile strength of the sandstone used for this study.

The only weakening mechanism applicable for sandstone samples subjected to cyclic wetting–drying treatment is the chemical and corrosive effect because the other four mechanisms mainly work under wet conditions. Aydan and Ulusay (2013) discovered that water content variations result in the swelling and shrinking of tuffs and are associated with the amount of clay in the tuffs. A few small hollows resulting from the dissolution of cementing material were observed on the surfaces of the sandstone

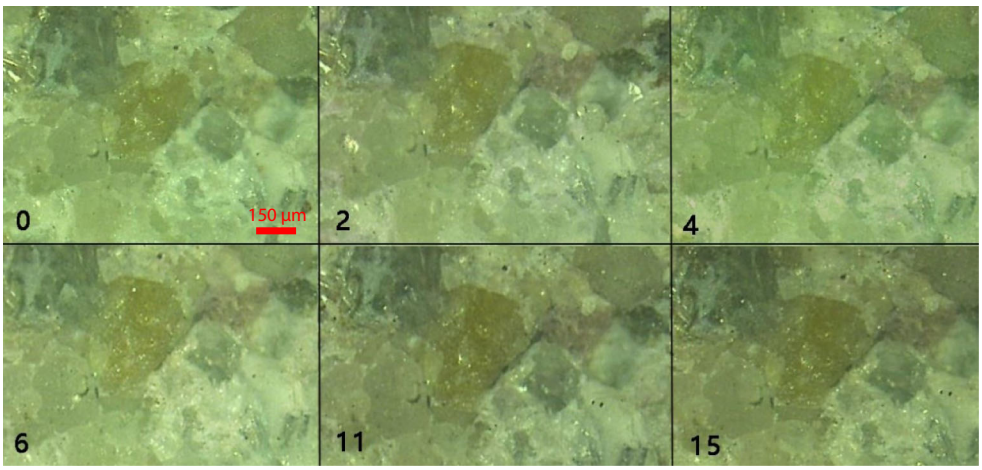


Fig. 4 Mineral dissolution of the sample A-5 during cyclic wetting–drying process. The numbers at the bottom-left corner of images indicate the number of wetting–drying cycles

samples used for this study, after a number of wetting–drying cycles (Fig. 4). However, features like this were sparse and could not be responsible for a significant reduction in tensile strength in the sandstone samples.

To what extent continuous wetting or cyclic wetting–drying can lower sandstone strength is strongly dependent on sandstone mineralogy and structures. The presence of soluble material like carbonate fragments, calcite and zeolite cement, or the presence of weak lamination planes, can make sandstones more sensitive to water weathering. For sandstones with a high quartz content and low clay mineral content, saturation-induced weakening may disappear after drying. For dams and railway or airport foundations, it is best to choose a sandstone with high quartz content because this type of aggregates is more stable under cyclic wetting–drying weathering conditions.

4.2 Outstanding Issues

As mentioned in Sect. 3.2, inter-sample variability is thought to be one possible reason for the lack of observable tensile strength reduction in the wetting–drying experiments. Another reason might be insufficient wetting–drying cycles to cause a significant reduction in strength. Hale and Shakoor (2003) did not observe unconfined compressive strength reductions even after 50 wetting–drying cycles. A further increase in the number of wetting–drying cycles should be attempted to see whether any strength reduction occurs in sandstones with a low content of clay minerals.

Deng et al. (2012) developed an apparatus that can apply water pressure to the immersed rock samples and their results showed that higher water pressures caused a larger reduction in compressive strength in sandstone samples that had undergone cyclic wetting–drying. In the literature (Table 1), samples are oven-dried at temperature above 100 °C, and the heated water might be able to mobilize soluble minerals toward the rock surface, which may weaken the rock. The experiments performed for this study avoided the temperature effects during cyclic wetting–drying treatment by drying the sandstone samples in air at room temperature, but the room humidity changed with time. It was shown, however, that the mass of the sandstone samples was not significantly affected by the changing room temperature and humidity during the cyclic wetting–drying.

5 Conclusions

The following conclusions can be drawn from this experimental study.

1. Sandstone tensile strength can be significantly lowered by wetting, but the saturation-induced reduction in tensile strength is probably not sensitive to the length of immersion. For sandstone with a low clay mineral content, the main water-weakening mechanisms are reductions in the fracture energy and the friction coefficient.
2. Cyclic wetting–drying does not necessarily reduce the sandstone's tensile strength permanently. This is because the only weakening mechanism during cyclic wetting–drying treatment is chemical and corrosive deterioration. For sandstone with a low of clay mineral content, the possible softening, expansion and dissolution of the clay minerals can play only a very minor role in weakening the sandstone. Any weakening due to saturation apparently disappears after drying.
3. Both dry and wet sandstone samples exhibit similar fracturing during Brazilian tensile testing. A primary

crack is initiated at the center of the sample and then propagated vertically along the loading direction. However, the presence of water substantially reduces the overall brittleness of the sandstone.

The above concluding remarks are only valid for the Chongqing sandstone investigated, a sandstone with a low clay mineral content. Whether the conclusions can be generalized and applied to other sandstones, like sandstone with a high clay mineral content, needs further study.

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References

Aydan Ö (2012) The inference of physico-mechanical properties of soft rocks and the evaluation of the effect of water content and weathering on their mechanical properties from needle penetration tests. In: The 46th US rock mechanics/geomechanics symposium, Chicago, 639

Aydan Ö, Ulusay R (2003) Geotechnical and geoenvironmental characteristics of man-made underground structures in Cappadocia, Turkey. Eng Geol 69:245–272

Aydan Ö, Ulusay R (2013) Geomechanical evaluation of Derinkuyu antique underground city and its implications in geoengineering. Rock Mech Rock Eng 46:731–754

Baechle GT, Weger RJ, Eberli GP, Massaferro JL, Sun YF (2005) Changes of shear moduli in carbonate rocks implications for Gassmann applicability. Leading Edge 24:507–510

Baud P, Zhu W, Wong T-F (2000) Failure mode and weakening effect of water on sandstone. J Geophys Res 105:16371–16389

Burshtein LS (1969) Effect of moisture on the strength and deformability of sandstone. Soviet Min 5:573–576

Cherblanc F, Berthonneau J, Bromblet P, Huon V (2016) Influence of water content on the mechanical behavior of limesonte: role of clay minerals content. Rock Mech Rock Eng 49:2033–2042

Colback PSB, Wild BL (1965) The influence of moisture content on the compressive strength of rock. In: Proceedings of the 3rd Canadian rock mechanics symposium, pp 65–83

Deng H, Li J, Zhu M, Wang K, Wang L, Deng C (2012) Experimental research on strength deterioration rules of sandstone under "saturation-air dry" circulation function. Rock Soil Mech 33:3306–3312 **(In Chinese)**

Duda M, Renner J (2013) The weakening effect of water on the brittle failure strength of sandstone. Geophys J Int 192:1091–1108

Hale PA, Shakoor A (2003) A laboratory investigation of the effects of cyclic heating and cooling, wetting and drying, and freezing and thawing on the compressive strength of selected sandstones. Environ Eng Geosci 9:117–130

Hawkins AB, McConnell BJ (1992) Sensitivity of sandstone strength and deformability to changes in moisture content. Q J Eng Geol 25:115–130

Hua W, Dong S, Li Y, Xu J, Wang Q (2015) The influence of cyclic wetting and drying on the fracture toughness of sandstone. Int J Rock Mech Min Sci 78:331–335

ISRM (1978) Suggested methods for determining tensile strength of rock materials. Int J Rock Mech Min Sci 15:99–103

Khanlari G, Abdilor Y (2015) Influence of wet-dry, freeze-thaw, and heat-cool cycles on the physical and mechanical properties of upper red sandstones in central Iran. Bull Eng Geol Environ 74:1287–1300

Li Z, Sheng Y, Reddish DJ (2005) Rock strength reduction and its potential environmental consequences as a result of groundwater rebound. In: The 9th International mine water congress. Oviedo, Asturias, Spain, pp 513–519

Lin ML, Jeng FS, Tsai LS, Huang TH (2005) Wetting weakening of tertiary sandstones - microscopic mechanism. Environ Geol 48:265–275

Özbek A (2014) Investigation of the effects of wetting–drying and freezing-thawing cycles on some physical and mechanical properties of selected ignimbrites. Bull Eng Geol Environ 73:595–609

Pimienta L, Fortin J, Guéguen Y (2014) Investigation of elastic weakening in limestone and sandstone samples from moisture adsorption. Geophys J Int 199:335–347

Reviron N, Reuschlé T, Bernard JD (2009) The brittle deformation regime of water-saturated siliceous sandstones. Geophys J Int 178:1766–1778

Sato A, Aydan Ö (2014) An X-Ray CT imaging of water absorption process of soft rocks. In: The 6th international conference on unsaturated soils. Sydney, Australia, pp 675–678

Shakoor A, Barefield EH (2009) Relationship between unconfined compressive strength and degree of saturation for selected sandstones. Environ Eng Geosci 15:29–40

Van Eeckhout EM (1976) The Mechanisms of strength reduction due to moisture in coal mine shales. Int J Rock Mech Min Sci Geomech Abstr 13:61–67

Vasarhelyi B (2003) Some observations regarding the strength and deformability of sandstones in case of dry and saturated conditions. Bull Eng Geol Environ 62:245–249

Vergara MR, Triantafyllidis T (2016) Influence of water content on the mechanical properties of an argillaceous swelling rock. Rock Mech Rock Eng 49:2555–2568

Wasantha PLP, Ranjith PG (2014) Water-weakening behavior of Hawkesbury sandstone in brittle regime. Eng Geol 178:91–101

Wong LNY, Jong MC (2014) Water saturation effects on the Brazilian tensile strength of gypsum and assessment of cracking processes using high-speed video. Rock Mech Rock Eng 47:1103–1105